Health claims and Insurance Intelligence Dashboard

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1) Objective and Business Goals

Objective

To deliver a centralized, interactive Power BI dashboard that provides end-to-end visibility into healthcare claims, insurance payments, provider performance, and member demographics — enabling faster, data-driven decisions.

Business Goals

Stakeholders (Insurance Executives, Claims Managers, Finance Teams, Provider Network Teams) aim to:

- Monitor total insurance payouts and patient payments
- Identify high-cost age groups and claim types
- Detect utilization patterns across providers
- Track financial performance and reimbursement efficiency
- Improve claim approval, cost control, and fraud monitoring
- Optimize provider network and member engagement strategies

Core Focus Areas:

1. **Claims Analytics**
2. **Provider & Member Insights**
3. **Financial Metrics**

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2) Data Sources and Data Model Overview

Data Sources

- Structured healthcare claims dataset (CSV/Excel/Database)
- Insurance reimbursement records
- Provider master data
- Member demographic data

(Dataset from Maven Analytics: healthcare.csv)

Data Model Design (Star Schema Recommended)

Fact Table:

Fact_Claims

- Claim_ID
- Provider_ID
- Member_ID
- Claim_Type
- Claim_Amount
- Insurance_Paid
- Patient_Payment
- Length_of_Stay
- Disease_Type
- Claim_Date

Dimension Tables:

Dim_Member

- Member_ID
- Age
- Age_Group
- Gender

Dim_Provider

- Provider_ID
- Provider_Name
- Provider_Type

Dim_Date

- Date
- Year
- Quarter
- Month

Dim_ClaimType

- Claim_Type (Commercial, Medicaid, Medicare, Employer Plan)

Relationships

- Fact_Claims → Dim_Member (Many-to-One)

- Fact_Claims → Dim_Provider (Many-to-One)
- Fact_Claims → Dim_Date (Many-to-One)
- Fact_Claims → Dim_ClaimType (Many-to-One)

Cross-filter direction: Single (to optimize performance)

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3) KPI and Measures List

Financial Metrics (Insurance Intelligence)

KPI	Definition
Total Insurance Provide	Total amount paid by insurance
Total Patient Payment	Out of pocket payments
Average Insurance Payment	Average reimbursement per claim
Average Patient Payment	Average Patient Contribution
Insurance Dependency Ratio	Revenue reliance on insurance

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4) Suggested Dashboard Structure

Page 1: Executive Overview

Top KPI Cards

- Total Insurance Provided
- Total Patient Payment
- Avg Insurance Payment
- Avg Patient Payment
- Avg Length of Stay

Visuals

- Stacked Column Chart: Claim Amount by Provider & Claim Type
- Pie Chart: Patient Count by Gender
- Line Chart: Insurance Paid vs Length of Stay
- Slicers: Age Group, Gender, Claim Type

Page 2: Claims & Financial Intelligence

Visuals

- Donut Chart: Claim Amount by Claim Type

- Bar Chart: Insurance Paid by Age Group
- Funnel Chart: Disease Count by Age Group
- Matrix: Provider vs Insurance & Patient Payment

Page 3 (Optional): Provider Deep Dive

Drill-through enabled:

- Provider performance trend
- Avg payment per claim
- Disease distribution per provider
- LOS comparison

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5) Data Refresh and Governance Plan

Data Refresh Strategy

- Dataset stored in Power BI Service
- Scheduled refresh: Daily or Weekly (based on business need)
- Gateway configured if using on-premise source

Governance & Security

Row-Level Security (RLS) Roles

- Finance Team → Full financial access
- Provider Manager → Provider-specific data only
- Executive → Aggregated view only

Best Practices

- Maintain certified dataset
- Version control via GitHub
- Data dictionary documentation
- Disable auto date/time

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6) Deployment and Rollout Plan

Phase 1 – Development

- Data cleaning in Power Query
- Data modeling
- DAX measure creation
- Visual design & UX testing

Phase 2 – Testing

- KPI validation with stakeholders
- Performance testing
- Security role testing

Phase 3 – Deployment

- Publish to Power BI Service
- Configure refresh
- Assign user roles
- Create App workspace

Phase 4 – Adoption

- Stakeholder demo
- Training session
- Feedback iteration

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7) Risks, Mitigations, and Prerequisites

Risk	Mitigation
Incomplete claim data	Data validation rules
Incorrect relationships	Star schema enforcement
Performance issues	Remove unused columns, optimize DAX
Security breach	Implement RLS and workspace roles
Misinterpretation of KPIs	Provide metric definitions

Prerequisites

- Clean and validated claims dataset
- Defined business KPIs
- Access to Power BI Service workspace
- Stakeholder alignment on metrics

Final Business Impact

This dashboard enables:

- Identification of high-cost age groups (e.g., 60+ seniors driving highest payouts)
- Visibility into insurance vs patient revenue dependency
- Detection of claim-type dominance (Medicare/Commercial trends)
- Provider-level performance benchmarking
- Strategic financial planning and cost optimization